dMAT
140

Mathematical Equations

Solve for every letter in each system.
47 easy, 46 moderate and 47 high-difficulty questions.

47	46	47
EASY	MODERATE	HIGH

Questions first. Answer key and worked solutions at the back.

Instructions and timed-practice tracker p. 2
Practice questions p. 3
Answer key p. 38
Worked solutions p. 40

Original practice questions following the formats and rule types in your uploaded dMAT General Academic Module preparation guide (07.07.2026). This is independent practice material, not an official examination paper.

Difficulty labels describe the intended practice progression; they are not official calibrated difficulty ratings.

Instructions

Solve each system so that every displayed equation is true. Find a value for every letter shown. Each letter represents an integer from 1 to 20, inclusive.

There is exactly one complete solution for each system. Different letters may have the same value when the equations require it. There are no multiple-choice options, matching the reference exercises.

Easy questions use two letters; moderate questions use three; high-difficulty questions use four. Addition, subtraction, multiplication and division follow the reference format. Apply multiplication and division before addition and subtraction.

Try substitution first: express several letters using one letter, solve for that letter, then recover the others. Check the full set of equations before marking your answer.

The solution pages number equations from top to bottom when explaining a route. These equation numbers are only references to the displayed lines.

The reference allows 25 minutes for 20 questions in this subtest and does not allow notes. First learn the methods, then use the seven 20-question blocks below for timed practice without working notes.

Timed-practice tracker

Block	Questions	Time	Correct	Retry
1	1-20	_____ min	_____ / 20	_____
2	21-40	_____ min	_____ / 20	_____
3	41-60	_____ min	_____ / 20	_____
4	61-80	_____ min	_____ / 20	_____
5	81-100	_____ min	_____ / 20	_____
6	101-120	_____ min	_____ / 20	_____
7	121-140	_____ min	_____ / 20	_____

Exercise 1 - Difficulty: easy

$$A - 11 = B$$

$$7 + B = 8$$

A = ____ B = ____

Exercise 2 - Difficulty: easy

$$B \div 4 = A$$

$$B - A = 3$$

A = ____ B = ____

Exercise 3 - Difficulty: easy

$$B + A = 16$$

$$3 \times B = A$$

A = ____ B = ____

Exercise 4 - Difficulty: easy

$$B - A = 13$$

$$B + A = 19$$

A = ____ B = ____

Exercise 5 - Difficulty: easy

$$A + 1 = B$$

$$2 \times A = B + 8$$

A = ____ B = ____

Exercise 6 - Difficulty: easy

$$4 \times B = 36$$

$$B + A = 25$$

A = ____ B = ____

Exercise 7 - Difficulty: easy

$$B \div 2 = A$$

$$21 - B = A$$

A = ____ B = ____

Exercise 8 - Difficulty: easy

$$A = B + 6$$

$$A + 3 \times B = 58$$

A = ____ B = ____

Exercise 9 - Difficulty: easy

$1 + A = 16$

$B + 14 = A$

A = ____ B = ____

Exercise 10 - Difficulty: easy

$A - B = 2$

$A \div 3 = B$

A = ____ B = ____

Exercise 11 - Difficulty: easy

$2 \times A = B$

$A + B = 27$

A = ____ B = ____

Exercise 12 - Difficulty: easy

$B - A = 11$

$B + A = 17$

A = ____ B = ____

Exercise 13 - Difficulty: easy

$3 \times B = A + 1$

$B + 1 = A$

A = ____ B = ____

Exercise 14 - Difficulty: easy

$B + A = 37$

$3 \times B = 60$

A = ____ B = ____

Exercise 15 - Difficulty: easy

$10 - A = B$

$A \div 4 = B$

A = ____ B = ____

Exercise 16 - Difficulty: easy

$A = B + 8$

$A + 3 \times B = 28$

A = ____ B = ____

Exercise 17 - Difficulty: easy

$$B + 10 = A$$

$$8 + A = 27$$

A = ____ B = ____

Exercise 18 - Difficulty: easy

$$A \div 4 = B$$

$$A - B = 15$$

A = ____ B = ____

Exercise 19 - Difficulty: easy

$$A + B = 12$$

$$2 \times A = B$$

A = ____ B = ____

Exercise 20 - Difficulty: easy

$$A + B = 29$$

$$A - B = 3$$

A = ____ B = ____

Exercise 21 - Difficulty: easy

$B + 9 = A$

$2 \times B = A + 2$

A = ____ B = ____

Exercise 22 - Difficulty: easy

$4 \times B = 60$

$B + A = 31$

A = ____ B = ____

Exercise 23 - Difficulty: easy

$20 - A = B$

$A \div 3 = B$

A = ____ B = ____

Exercise 24 - Difficulty: easy

$B = A + 14$

$B + 3 \times A = 30$

A = ____ B = ____

Exercise 25 - Difficulty: easy

$$A - 7 = B$$
$$4 + B = 10$$

A = ____ B = ____

Exercise 26 - Difficulty: easy

$$B \div 4 = A$$
$$B - A = 6$$

A = ____ B = ____

Exercise 27 - Difficulty: easy

$$A + B = 25$$
$$4 \times A = B$$

A = ____ B = ____

Exercise 28 - Difficulty: easy

$$A + B = 31$$
$$A - B = 1$$

A = ____ B = ____

Exercise 29 - Difficulty: easy

$3 \times A = B + 5$

$A + 11 = B$

A = ____ B = ____

Exercise 30 - Difficulty: easy

$A + B = 8$

$4 \times A = 20$

A = ____ B = ____

Exercise 31 - Difficulty: easy

$16 - A = B$

$A \div 3 = B$

A = ____ B = ____

Exercise 32 - Difficulty: easy

$B = A + 3$

$B + 2 \times A = 18$

A = ____ B = ____

Exercise 33 - Difficulty: easy

$4 + A = 13$

$B + 4 = A$

A = ____ B = ____

Exercise 34 - Difficulty: easy

$A - B = 8$

$A \div 3 = B$

A = ____ B = ____

Exercise 35 - Difficulty: easy

$2 \times B = A$

$B + A = 15$

A = ____ B = ____

Exercise 36 - Difficulty: easy

$B - A = 1$

$B + A = 17$

A = ____ B = ____

Exercise 37 - Difficulty: easy

$4 \times A = B + 7$

$A + 11 = B$

A = ____ B = ____

Exercise 38 - Difficulty: easy

$A + B = 24$

$2 \times A = 26$

A = ____ B = ____

Exercise 39 - Difficulty: easy

$18 - B = A$

$B \div 2 = A$

A = ____ B = ____

Exercise 40 - Difficulty: easy

$B + 4 \times A = 79$

$B = A + 4$

A = ____ B = ____

Exercise 41 - Difficulty: easy

$1 + B = 4$

$A - 3 = B$

A = ____ B = ____

Exercise 42 - Difficulty: easy

$A - B = 4$

$A \div 2 = B$

A = ____ B = ____

Exercise 43 - Difficulty: easy

$A + B = 16$

$3 \times A = B$

A = ____ B = ____

Exercise 44 - Difficulty: easy

$B + A = 39$

$B - A = 1$

A = ____ B = ____

Exercise 45 - Difficulty: easy

$$A + 2 = B$$
$$2 \times A = B + 14$$

A = ____ B = ____

Exercise 46 - Difficulty: easy

$$A + B = 18$$
$$3 \times A = 45$$

A = ____ B = ____

Exercise 47 - Difficulty: easy

$$12 - A = B$$
$$A \div 3 = B$$

A = ____ B = ____

Exercise 48 - Difficulty: moderate

$$2 \times A = C$$
$$1 \times C + 1 \times A = B$$
$$C + A = 6$$

A = ____ B = ____ C = ____

Exercise 49 - Difficulty: moderate

$20 - B = C$

$2 \times C = A$

$B \div 4 = C$

A = ____ B = ____ C = ____

Exercise 50 - Difficulty: moderate

$A = C + 6$

$A + C + B = 22$

$C = B + 5$

A = ____ B = ____ C = ____

Exercise 51 - Difficulty: moderate

$A + C = 24$

$A + B = 14$

$B + C = 22$

A = ____ B = ____ C = ____

Exercise 52 - Difficulty: moderate

$B = 3 \times C$

$C + A = 11$

$A = B - 5$

A = ____ B = ____ C = ____

Exercise 53 - Difficulty: moderate

$C + A = 4$

$A = B + 1$

$B = C \div 2$

A = ____ B = ____ C = ____

Exercise 54 - Difficulty: moderate

$A = C - B$

$2 \times C + B = 23$

$A = 4$

A = ____ B = ____ C = ____

Exercise 55 - Difficulty: moderate

$B = 3 \times C$

$B + A = 11$

$C = A - 7$

A = ____ B = ____ C = ____

Exercise 56 - Difficulty: moderate

$2 \times B + 2 \times C = A$

$4 \times C = B$

$B + C = 10$

A = ____ B = ____ C = ____

Exercise 57 - Difficulty: moderate

$2 \times B = C$

$A \div 4 = B$

$15 - A = B$

A = ____ B = ____ C = ____

Exercise 58 - Difficulty: moderate

$B = A + 6$

$B + A + C = 41$

$A = C + 4$

A = ____ B = ____ C = ____

Exercise 59 - Difficulty: moderate

$A + B = 24$

$A + C = 11$

$B + C = 23$

A = ____ B = ____ C = ____

Exercise 60 - Difficulty: moderate

$A = C - 2$

$C = 2 \times B$

$B + A = 4$

A = ____ B = ____ C = ____

Exercise 61 - Difficulty: moderate

$C = A + 3$

$A = B \div 2$

$B + C = 21$

A = ___ B = ___ C = ___

Exercise 62 - Difficulty: moderate

$4 \times A + B = 52$

$C = A - B$

$C = 8$

A = ___ B = ___ C = ___

Exercise 63 - Difficulty: moderate

$A = 2 \times C$

$A + B = 19$

$C = B - 7$

A = ___ B = ___ C = ___

Exercise 64 - Difficulty: moderate

$C + A = 10$

$4 \times A = C$

$2 \times C + 1 \times A = B$

A = ___ B = ___ C = ___

Exercise 65 - Difficulty: moderate

$12 - C = B$

$C \div 2 = B$

$3 \times B = A$

A = ____ B = ____ C = ____

Exercise 66 - Difficulty: moderate

$C + A + B = 42$

$A = B + 4$

$C = A + 4$

A = ____ B = ____ C = ____

Exercise 67 - Difficulty: moderate

$A + B = 19$

$C + A = 26$

$C + B = 17$

A = ____ B = ____ C = ____

Exercise 68 - Difficulty: moderate

$A = 2 \times C$

$C + B = 13$

$B = A - 2$

A = ____ B = ____ C = ____

Exercise 69 - Difficulty: moderate

$$B + C = 17$$
$$C = A + 7$$
$$A = B \div 4$$

A = ___ B = ___ C = ___

Exercise 70 - Difficulty: moderate

$$4 \times C + A = 79$$
$$B = 1$$
$$B = C - A$$

A = ___ B = ___ C = ___

Exercise 71 - Difficulty: moderate

$$B + C = 6$$
$$B = 2 \times A$$
$$A = C - 3$$

A = ___ B = ___ C = ___

Exercise 72 - Difficulty: moderate

$$C + B = 5$$
$$1 \times C + 1 \times B = A$$
$$4 \times B = C$$

A = ___ B = ___ C = ___

Exercise 73 - Difficulty: moderate

$A \div 3 = C$

$2 \times C = B$

$12 - A = C$

A = ____ B = ____ C = ____

Exercise 74 - Difficulty: moderate

$B = A + 1$

$B + A + C = 23$

$A = C + 5$

A = ____ B = ____ C = ____

Exercise 75 - Difficulty: moderate

$C + B = 36$

$A + B = 28$

$A + C = 30$

A = ____ B = ____ C = ____

Exercise 76 - Difficulty: moderate

$B + C = 7$

$A = 4 \times B$

$C = A - 3$

A = ____ B = ____ C = ____

Exercise 77 - Difficulty: moderate

$A + C = 14$

$B = A \div 2$

$C = B + 2$

A = ___ B = ___ C = ___

Exercise 78 - Difficulty: moderate

$C = 10$

$C = A - B$

$4 \times A + B = 50$

A = ___ B = ___ C = ___

Exercise 79 - Difficulty: moderate

$A = 4 \times C$

$A + B = 19$

$C = B - 4$

A = ___ B = ___ C = ___

Exercise 80 - Difficulty: moderate

$1 \times B + 2 \times C = A$

$B + C = 3$

$2 \times C = B$

A = ___ B = ___ C = ___

Exercise 81 - Difficulty: moderate

$16 - C = B$

$C \div 3 = B$

$2 \times B = A$

A = ____ B = ____ C = ____

Exercise 82 - Difficulty: moderate

$B = A + 2$

$B + A + C = 37$

$A = C + 4$

A = ____ B = ____ C = ____

Exercise 83 - Difficulty: moderate

$B + C = 14$

$A + C = 15$

$B + A = 9$

A = ____ B = ____ C = ____

Exercise 84 - Difficulty: moderate

$C = A - 2$

$A = 2 \times B$

$B + C = 13$

A = ____ B = ____ C = ____

Exercise 85 - Difficulty: moderate

$$B = C + 1$$
$$C = A \div 4$$
$$A + B = 6$$

A = ____ B = ____ C = ____

Exercise 86 - Difficulty: moderate

$$C = A - B$$
$$C = 6$$
$$2 \times A + B = 36$$

A = ____ B = ____ C = ____

Exercise 87 - Difficulty: moderate

$$B + A = 31$$
$$C = A - 6$$
$$B = 4 \times C$$

A = ____ B = ____ C = ____

Exercise 88 - Difficulty: moderate

$$4 \times A = B$$
$$B + A = 15$$
$$1 \times B + 1 \times A = C$$

A = ____ B = ____ C = ____

Exercise 89 - Difficulty: moderate

$6 - A = B$

$2 \times B = C$

$A \div 2 = B$

A = ____ B = ____ C = ____

Exercise 90 - Difficulty: moderate

$A = B + 6$

$A + B + C = 40$

$B = C + 5$

A = ____ B = ____ C = ____

Exercise 91 - Difficulty: moderate

$C + B = 30$

$A + C = 22$

$A + B = 28$

A = ____ B = ____ C = ____

Exercise 92 - Difficulty: moderate

$C + A = 14$

$B = 4 \times C$

$A = B - 6$

A = ____ B = ____ C = ____

Exercise 93 - Difficulty: moderate

$B = C \div 4$

$A = B + 6$

$C + A = 16$

A = ___ B = ___ C = ___

Exercise 94 - Difficulty: high

$4 \times C = A$

$3 \times C = D$

$D - C + A - B = 9$

$6 + C = B$

A = ___ B = ___ C = ___ D = ___

Exercise 95 - Difficulty: high

$C + D - B = 19$

$3 \times C - 1 = A$

$11 - C = B$

$4 \times C = D$

A = ___ B = ___ C = ___ D = ___

Exercise 96 - Difficulty: high

$D = 2 \times C - 1$

$A = 2 \times B - 3$

$C = A + 1$

$D - C + A + B = 4$

A = ___ B = ___ C = ___ D = ___

Exercise 97 - Difficulty: high

$B + A = 6$

$C + B = 16$

$D = 3 \times A$

$D + C = 18$

A = ____ B = ____ C = ____ D = ____

Exercise 98 - Difficulty: high

$A \div 3 + D = B$

$D = C + 2$

$C = B - D$

$A + D + B + C = 32$

A = ____ B = ____ C = ____ D = ____

Exercise 99 - Difficulty: high

$5 \times A - D = B$

$A + C = 13$

$D + B = 15$

$C = B - A$

A = ____ B = ____ C = ____ D = ____

Exercise 100 - Difficulty: high

$C + B + D = 24$

$D = 2 \times A$

$C - B = 6$

$B = A + 3$

A = ____ B = ____ C = ____ D = ____

Exercise 101 - Difficulty: high

$B = 5 \times C - 5$

$D = 4 \times C - 4$

$A = C + 5$

$B + C = A + D - 4$

A = ____ B = ____ C = ____ D = ____

Exercise 102 - Difficulty: high

$C - A + D - B = 21$

$4 \times A = D$

$9 + A = B$

$4 \times A = C$

A = ____ B = ____ C = ____ D = ____

Exercise 103 - Difficulty: high

$5 \times C - 1 = B$

$10 - C = A$

$5 \times C = D$

$C + D - A = 4$

A = ____ B = ____ C = ____ D = ____

Exercise 104 - Difficulty: high

$A - B + C + D = 19$

$A = 3 \times B - 4$

$C = 3 \times D - 7$

$B = C + 2$

A = ____ B = ____ C = ____ D = ____

Exercise 105 - Difficulty: high

$D + C = 12$

$B + D = 11$

$A = 2 \times C$

$A + B = 8$

A = ____ B = ____ C = ____ D = ____

Exercise 106 - Difficulty: high

$A + D + C + B = 29$

$A \div 5 + D = C$

$D = B + 1$

$B = C - D$

A = ____ B = ____ C = ____ D = ____

Exercise 107 - Difficulty: high

$D + B = 16$

$4 \times D - A = C$

$B = C - D$

$A + C = 24$

A = ____ B = ____ C = ____ D = ____

Exercise 108 - Difficulty: high

$A = 3 \times D$

$B = D + 3$

$C + B + A = 33$

$C - B = 2$

A = ____ B = ____ C = ____ D = ____

Exercise 109 - Difficulty: high

$A = 4 \times C - 3$

$D = C + 2$

$B = 5 \times C - 10$

$A + C = D + B$

A = ____ B = ____ C = ____ D = ____

Exercise 110 - Difficulty: high

$2 \times A = C$

$3 + A = D$

$C - A + B - D = 13$

$4 \times A = B$

A = ____ B = ____ C = ____ D = ____

Exercise 111 - Difficulty: high

$11 - B = C$

$5 \times B = D$

$5 \times B - 1 = A$

$B + D - C = 3$

A = ____ B = ____ C = ____ D = ____

Exercise 112 - Difficulty: high

$C = 5 \times B - 6$

$A - D + C + B = 10$

$D = C + 9$

$A = 2 \times D - 9$

A = ____ B = ____ C = ____ D = ____

Exercise 113 - Difficulty: high

$C + A = 24$

$B = 4 \times D$

$B + C = 15$

$A + D = 14$

A = ____ B = ____ C = ____ D = ____

Exercise 114 - Difficulty: high

$B \div 2 + C = A$

$B + C + A + D = 34$

$C = D + 5$

$D = A - C$

A = ____ B = ____ C = ____ D = ____

Exercise 115 - Difficulty: high

$D + B = 20$

$B = C - D$

$A + C = 24$

$4 \times D - A = C$

A = ____ B = ____ C = ____ D = ____

Exercise 116 - Difficulty: high

$C = 4 \times A$

$B = A + 5$

$D - B = 1$

$D + B + C = 23$

A = ____ B = ____ C = ____ D = ____

Exercise 117 - Difficulty: high

$D = 2 \times B - 7$

$A = B + 5$

$C = 4 \times B - 5$

$D + B = A + C - 19$

A = ____ B = ____ C = ____ D = ____

Exercise 118 - Difficulty: high

$D - A + B - C = 14$

$4 \times A = D$

$3 \times A = B$

$1 + A = C$

A = ____ B = ____ C = ____ D = ____

Exercise 119 - Difficulty: high

$C + A - B = 6$

$2 \times C - 1 = D$

$9 - C = B$

$3 \times C = A$

A = ____ B = ____ C = ____ D = ____

Exercise 120 - Difficulty: high

$A = 3 \times D - 3$

$D = C + 1$

$C = 3 \times B - 4$

$A - D + C + B = 7$

A = ____ B = ____ C = ____ D = ____

Exercise 121 - Difficulty: high

D + A = 10

C + D = 11

B + C = 16

B = 4 × A

A = ____ B = ____ C = ____ D = ____

Exercise 122 - Difficulty: high

A = C - B

D ÷ 3 + B = C

D + B + C + A = 28

B = A + 7

A = ____ B = ____ C = ____ D = ____

Exercise 123 - Difficulty: high

D = B - C

C + D = 13

2 × C - A = B

A + B = 20

A = ____ B = ____ C = ____ D = ____

Exercise 124 - Difficulty: high

A = D + 1

B = 3 × D

C + A + B = 24

C - A = 7

A = ____ B = ____ C = ____ D = ____

Exercise 125 - Difficulty: high

$B = 3 \times A - 4$

$D = A + 2$

$C = 4 \times A - 5$

$C + A = D + B + 2$

A = ___ B = ___ C = ___ D = ___

Exercise 126 - Difficulty: high

$3 \times D = A$

$3 \times D = C$

$C - D + A - B = 9$

$3 + D = B$

A = ___ B = ___ C = ___ D = ___

Exercise 127 - Difficulty: high

$4 \times C = B$

$3 \times C - 1 = D$

$12 - C = A$

$C + B - A = 6$

A = ___ B = ___ C = ___ D = ___

Exercise 128 - Difficulty: high

$D = 4 \times B - 2$

$A = 3 \times C - 5$

$A - C + D + B = 6$

$C = D + 2$

A = ___ B = ___ C = ___ D = ___

Exercise 129 - Difficulty: high

$D = 4 \times B$

$C + B = 15$

$A + C = 21$

$D + A = 31$

A = ____ B = ____ C = ____ D = ____

Exercise 130 - Difficulty: high

$B + C + A + D = 31$

$D = A - C$

$C = D + 5$

$B \div 3 + C = A$

A = ____ B = ____ C = ____ D = ____

Exercise 131 - Difficulty: high

$B = D - C$

$A + D = 16$

$C + B = 8$

$4 \times C - A = D$

A = ____ B = ____ C = ____ D = ____

Exercise 132 - Difficulty: high

$C = A + 2$

$D = 4 \times A$

$B - C = 8$

$B + C + D = 30$

A = ____ B = ____ C = ____ D = ____

Exercise 133 - Difficulty: high

$B = 4 \times C - 3$

$A = 5 \times C - 6$

$A + C = D + B - 3$

$D = C + 3$

A = ____ B = ____ C = ____ D = ____

Exercise 134 - Difficulty: high

$6 + A = D$

$B - A + C - D = 6$

$4 \times A = C$

$2 \times A = B$

A = ____ B = ____ C = ____ D = ____

Exercise 135 - Difficulty: high

$2 \times A - 1 = D$

$2 \times A = C$

$13 - A = B$

$A + C - B = 3$

A = ____ B = ____ C = ____ D = ____

Exercise 136 - Difficulty: high

$A = 2 \times B - 7$

$A - B + D + C = 8$

$B = D + 2$

$D = 3 \times C - 4$

A = ____ B = ____ C = ____ D = ____

Exercise 137 - Difficulty: high

$B + C = 11$

$C + A = 6$

$D + B = 8$

$D = 2 \times A$

A = ____ B = ____ C = ____ D = ____

Exercise 138 - Difficulty: high

$D = A + 4$

$C + D + B + A = 32$

$C \div 4 + D = B$

$A = B - D$

A = ____ B = ____ C = ____ D = ____

Exercise 139 - Difficulty: high

$5 \times B - D = A$

$D + A = 15$

$C = A - B$

$B + C = 14$

A = ____ B = ____ C = ____ D = ____

Exercise 140 - Difficulty: high

$C - D = 4$

$D = A + 2$

$B = 3 \times A$

$C + D + B = 33$

A = ____ B = ____ C = ____ D = ____